EXP 6 : Simulation of SISO and MIMO Systems: BER and Beamforming Performance Analysis

MATLAB CODE:

```
clc;
clear;
close all;

%% SISO AND MIMO SYSTEM SIMULATION
% BER and Beamforming Performance Analysis

SNR_dB = 0:2:20;
Nbits = 100000;

BER_SISO = zeros(size(SNR_dB));
BER_MIMO = zeros(size(SNR_dB));
BER_BF = zeros(size(SNR_dB));

%% 1. BER SIMULATION

for k = 1:length(SNR_dB)

    SNR = 10^(SNR_dB(k)/10);
    N0 = 1/SNR;

    % Generate BPSK data
    bits = randi([0 1],1,Nbits);
    symbols = 2*bits-1;

    %% SISO SYSTEM (1 TRANSMIT, 1 RECEIVE)

    h = (randn(1,Nbits)+1i*randn(1,Nbits))/sqrt(2);

    noise = sqrt(N0/2)*(randn(1,Nbits)+ ...
                    1i*randn(1,Nbits));

    received = h.*symbols + noise;

    equalized = received.*conj(h);

    detected = real(equalized)>=0;

    BER_SISO(k) = sum(detected~=bits)/Nbits;

    %% 2x1 MIMO BEAMFORMING (MRT)

    h1 = (randn(1,Nbits)+1i*randn(1,Nbits))/sqrt(2);
    h2 = (randn(1,Nbits)+1i*randn(1,Nbits))/sqrt(2);
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channelNorm = sqrt(abs(h1).^2+abs(h2).^2);

% Transmit beamforming weights
w1 = conj(h1)./channelNorm;
w2 = conj(h2)./channelNorm;

% Received signal
noise = sqrt(N0/2)*(randn(1,Nbits)+ ...
                1i*randn(1,Nbits));

receivedBF = h1.*w1.*symbols + ...
          h2.*w2.*symbols + noise;

equalizedBF = receivedBF./channelNorm;

detectedBF = real(equalizedBF)>=0;

BER_BF(k) = sum(detectedBF~=bits)/Nbits;

%% 2x2 MIMO ALAMOUTI SYSTEM

% Two symbols per transmission block
s1 = symbols(1:2:end);
s2 = symbols(2:2:end);

Nblocks = length(s1);

% Four independent Rayleigh fading channels
h11 = (randn(1,Nblocks)+1i*randn(1,Nblocks))/sqrt(2);
h12 = (randn(1,Nblocks)+1i*randn(1,Nblocks))/sqrt(2);

h21 = (randn(1,Nblocks)+1i*randn(1,Nblocks))/sqrt(2);
h22 = (randn(1,Nblocks)+1i*randn(1,Nblocks))/sqrt(2);

% Time slot 1
n11 = sqrt(N0/2)*(randn(1,Nblocks)+1i*randn(1,Nblocks));
n21 = sqrt(N0/2)*(randn(1,Nblocks)+1i*randn(1,Nblocks));

y11 = (h11.*s1+h12.*s2)/sqrt(2)+n11;
y21 = (h21.*s1+h22.*s2)/sqrt(2)+n21;

% Time slot 2
n12 = sqrt(N0/2)*(randn(1,Nblocks)+1i*randn(1,Nblocks));
n22 = sqrt(N0/2)*(randn(1,Nblocks)+1i*randn(1,Nblocks));

y12 = (-h11.*s2+h12.*s1)/sqrt(2)+n12;
y22 = (-h21.*s2+h22.*s1)/sqrt(2)+n22;

% Alamouti decoding
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```
    H = abs(h11).^2+abs(h12).^2+ ...
        abs(h21).^2+abs(h22).^2;

    z1 = conj(h11).*y11+h12.*conj(y12)+ ...
         conj(h21).*y21+h22.*conj(y22);

    z2 = conj(h12).*y11-h11.*conj(y12)+ ...
         conj(h22).*y21-h21.*conj(y22);

    % Normalize received signals
    s1_hat = z1./(H/sqrt(2));
    s2_hat = z2./(H/sqrt(2));

    detected1 = real(s1_hat)>=0;
    detected2 = real(s2_hat)>=0;

    detectedMIMO = zeros(1,Nbits);

    detectedMIMO(1:2:end) = detected1;
    detectedMIMO(2:2:end) = detected2;

    BER_MIMO(k) = sum(detectedMIMO~=bits)/Nbits;

end

%% 2. BER COMPARISON GRAPH

figure('Name','BER Comparison');

semilogy(SNR_dB,max(BER_SISO,1/Nbits), ...
    'r-o','LineWidth',2);

hold on;

semilogy(SNR_dB,max(BER_MIMO,1/Nbits), ...
    'b-s','LineWidth',2);

semilogy(SNR_dB,max(BER_BF,1/Nbits), ...
    'g-^','LineWidth',2);

xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');

title('SISO and MIMO BER Performance');

legend('SISO 1x1','MIMO 2x2 Alamouti', ...
       'Beamforming 2x1','Location','southwest');

grid on;
```

```
%% 3. BEAMFORMING ANTENNA ARRAY

Nant = 8;                % Number of antenna elements
d = 0.5;                 % Antenna spacing in wavelengths
theta0 = 30;             % Desired beam direction

theta = -90:0.1:90;

% Antenna array steering vector
steering = exp(1i*2*pi*d*(0:Nant-1)'*sind(theta));

% Beamforming weights
w = exp(1i*2*pi*d*(0:Nant-1)'*sind(theta0))/sqrt(Nant);

% Calculate array factor
beamPattern = abs(w'*steering);

% Normalize pattern
beamPattern = beamPattern/max(beamPattern);

beamPattern_dB = 20*log10(max(beamPattern,1e-4));

%% 4. BEAM PATTERN GRAPH

figure('Name','Beamforming Pattern');

plot(theta,beamPattern_dB,'b','LineWidth',2);

hold on;

xline(theta0,'r--','Desired Direction');

xlabel('Angle (Degrees)');
ylabel('Normalized Gain (dB)');

title('8 Element MIMO Beamforming Pattern');

ylim([-40 2]);
xlim([-90 90]);

grid on;

%% 5. BEAMFORMING GAIN COMPARISON

figure('Name','Beamforming Gain');

bar([1,Nant]);

set(gca,'XTickLabel',{'Single Antenna','8 Antenna Array'});
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```
ylabel('Relative Power Gain (dB)');

title('Beamforming Array Gain Comparison');

grid on;

%% 6. FINAL BER RESULTS

fprintf('\n--- SISO AND MIMO PERFORMANCE ---\n');

fprintf('\nSNR(dB)     SISO BER      MIMO BER      BF BER\n');

for k = 1:length(SNR_dB)

   fprintf('%5d      %.5e    %.5e    %.5e\n', ...
      SNR_dB(k),BER_SISO(k), ...
      BER_MIMO(k),BER_BF(k));

end

%% 7. FINAL COMPARISON TABLE

Results = table( ...
   SNR_dB', ...
   BER_SISO', ...
   BER_MIMO', ...
   BER_BF', ...
   'VariableNames', ...
   {'SNR_dB','SISO_BER','MIMO_2x2_BER','Beamforming_BER'});

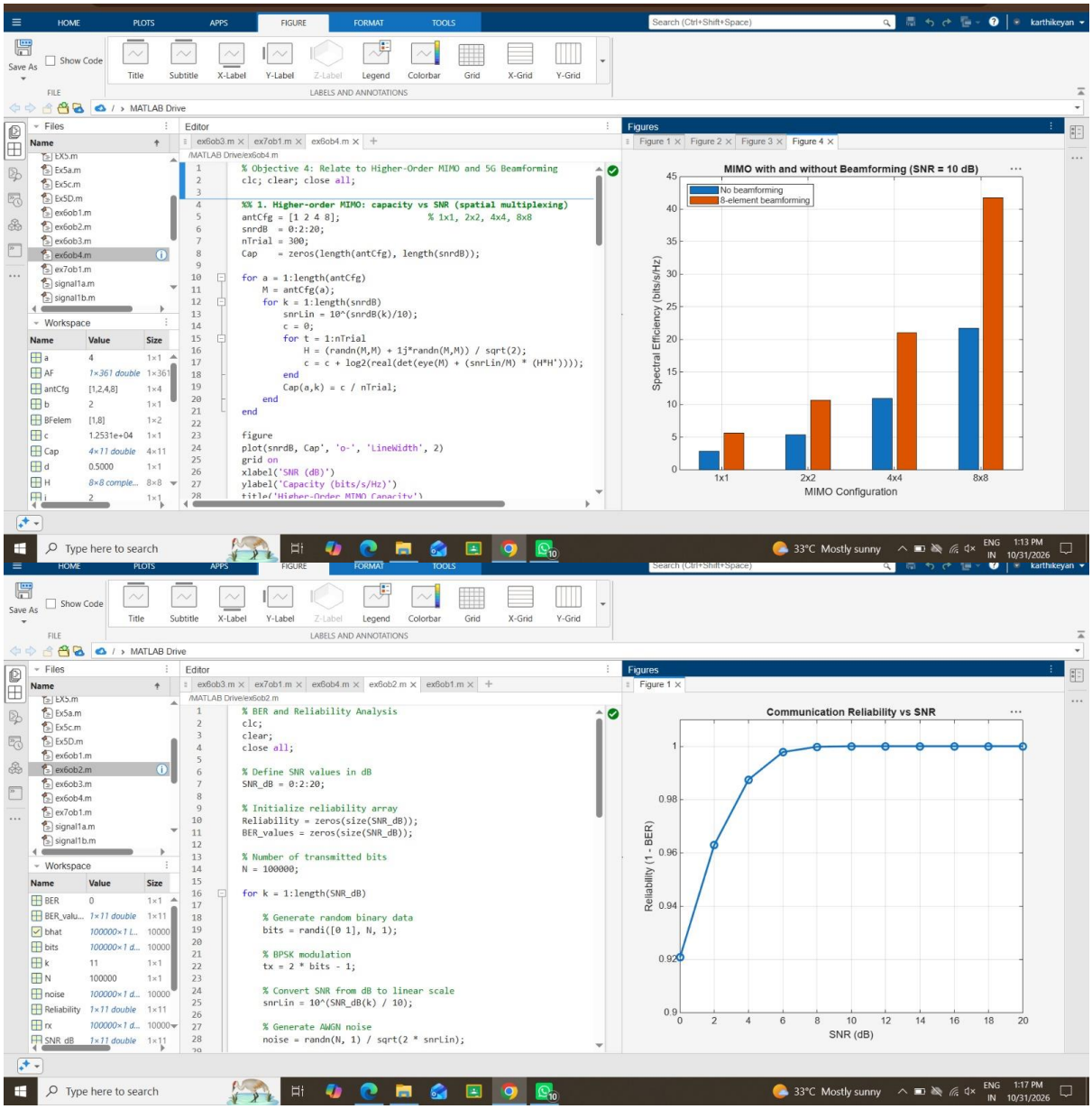
disp(' ');
disp('--- FINAL COMPARISON TABLE ---');

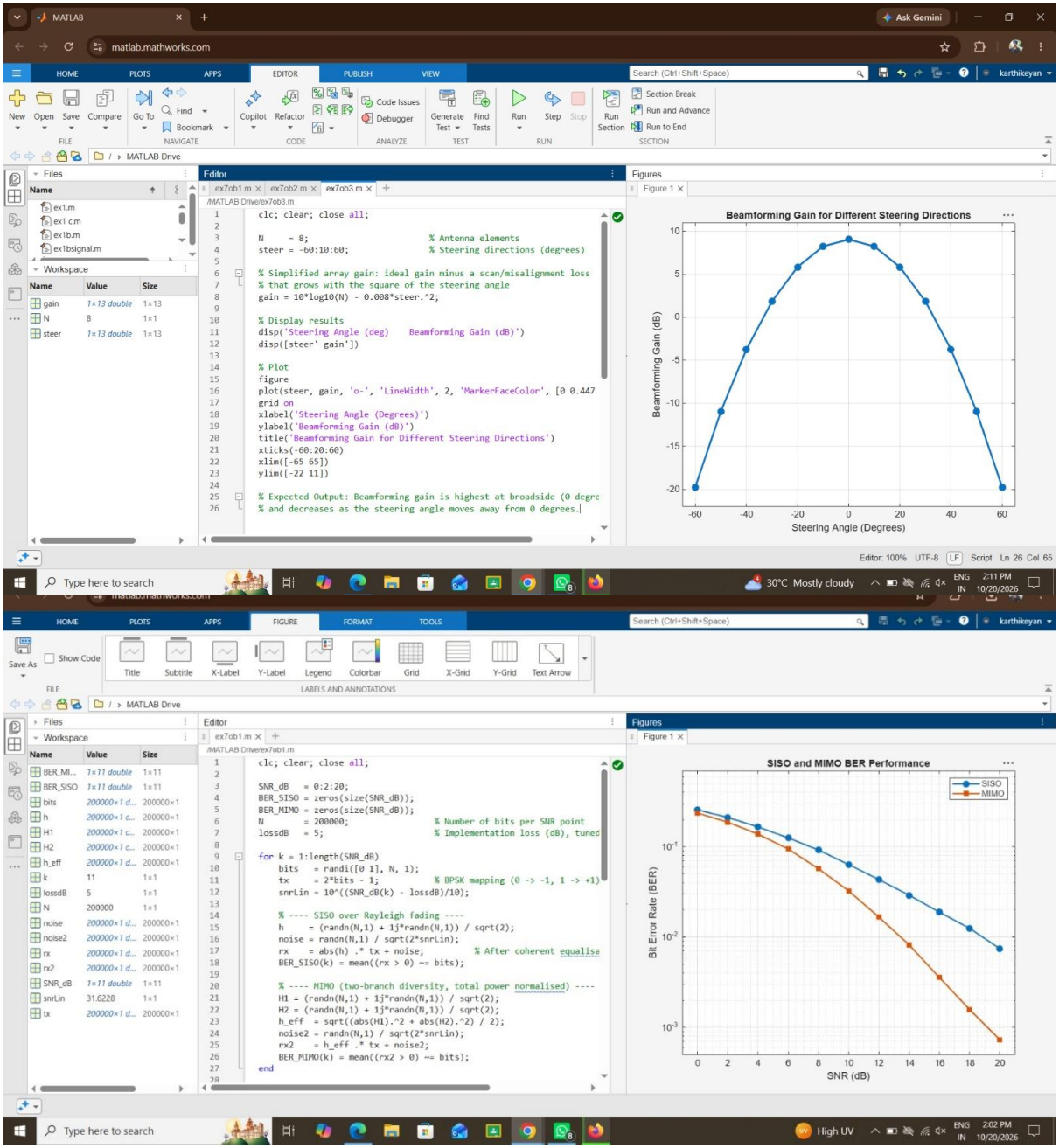
disp(Results);

fprintf('\nNumber of Antennas: %d\n',Nant);
fprintf('Beamforming Direction: %d degrees\n',theta0);

fprintf('\nSimulation Completed Successfully!\n');
```

OUTPUT:


MIMO with and without Beamforming (SNR = 10 dB)
No beamforming
8-element beamforming
Spectral Efficiency (bits/s/Hz)
MIMO Configuration
1x1
2x2
4x4
8x8
Communication Reliability vs SNR
Reliability (1 - BER)
SNR (dB)


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ex7ob1.m
ex7ob2.m
ex7ob3.m
clc; clear; close all;
N = 8; % Antenna elements
steer = -60:10:60; % Steering directions (degrees)
% Simplified array gain: ideal gain minus a scan/misalignment loss
% that grows with the square of the steering angle
gain = 10*log10(N) - 0.008*steer.^2;
% Display results
disp('Steering Angle (deg) Beamforming Gain (dB)')
disp([steer' gain'])
% Plot
figure
plot(steer, gain, 'o-', 'LineWidth', 2, 'MarkerFaceColor', [0 0.447
grid on
xlabel('Steering Angle (Degrees)')
ylabel('Beamforming Gain (dB)')
title('Beamforming Gain for Different Steering Directions')
xticks(-60:20:60)
xlim([-65 65])
ylim([-22 11])
% Expected Output: Beamforming gain is highest at broadside (0 degre
% and decreases as the steering angle moves away from 0 degrees.
Figures
Figure 1
Beamforming Gain for Different Steering Directions
Beamforming Gain (dB)
Steering Angle (Degrees)
Editor: 100% UTF-8 LF Script Ln 26 Col 65
Type here to search
30°C Mostly cloudy
ENG IN 2:11 PM 10/20/2026
FIGURE
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Title
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Y-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
Text Arrow
FILE
LABELS AND ANNOTATIONS
ex7ob1.m
clc; clear; close all;
SNR_dB = 0:2:20;
BER_SISO = zeros(size(SNR_dB));
BER_MIMO = zeros(size(SNR_dB));
N = 200000; % Number of bits per SNR point
lossdB = 5; % Implementation loss (dB), tuned
for k = 1:length(SNR_dB)
bits = randi([0 1], N, 1);
tx = 2*bits - 1; % BPSK mapping (0 -> -1, 1 -> +1)
snrLin = 10^((SNR_dB(k) - lossdB)/10);
% ---- SISO over Rayleigh fading ----
h = (randn(N,1) + 1j*randn(N,1)) / sqrt(2);
noise = randn(N,1) / sqrt(2*snrLin);
rx = abs(h) .* tx + noise; % After coherent equalise
BER_SISO(k) = mean((rx > 0) ~= bits);
% ---- MIMO (two-branch diversity, total power normalised) ----
H1 = (randn(N,1) + 1j*randn(N,1)) / sqrt(2);
H2 = (randn(N,1) + 1j*randn(N,1)) / sqrt(2);
h_eff = sqrt(abs(H1).^2 + abs(H2).^2) / 2);
noise2 = randn(N,1) / sqrt(2*snrLin);
rx2 = h_eff .* tx + noise2;
BER_MIMO(k) = mean((rx2 > 0) ~= bits);
end
SISO and MIMO BER Performance
SISO
MIMO
Bit Error Rate (BER)
SNR (dB)
High UV
ENG IN 2:02 PM 10/20/2026